

1. Administrative & Study Identification

Full Study Title: Clinical Study to Evaluate the Impact of the Accu-Chek SmartGuide CGM Solution on the Mean Change in Time in Range Compared With Self-Monitoring of Blood Glucose in Participants With Type 1 and Type 2 Diabetes Mellitus **Short Title/Acronym:** Accu-Chek SmartGuide CGM TIR Study **Protocol Number:** DC000129 **NCT Number:** NCT06704672 **Sponsor Name:** Hoffmann-La Roche **Investigational Product:** Accu-Chek SmartGuide CGM Solution (Sensor, SmartGuide App, SmartGuide Predict App) **Phase of Development:** N/A (Device/Quality Improvement Trial) **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the impact of the Accu-Chek SmartGuide CGM solution on the mean change in overall time in range (TIR) of blood glucose concentrations of 70-180 mg/dl compared with self-monitoring of blood glucose (SMBG). **Secondary Objectives:** To evaluate the incidence of Level 1 and Level 2 hypoglycemia, user satisfaction with the SmartGuide predictive tools, safety/tolerability of the sensor adhesive, and overall HbA1c reduction.

2.2 Background & Rationale To optimize glycemic control and increase the time in range (TIR) for adults with Type 1 Diabetes (T1D) or insulin-dependent Type 2 Diabetes (T2D) on a multiple daily injection (MDI) regimen. Continuous Glucose Monitoring (CGM) solutions provide real-time visibility into interstitial glucose fluctuations and predictive trends, potentially bridging the gap where traditional, episodic SMBG fails to detect silent hyper- and hypoglycemic events.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Device Study) **Control Type:** Active Comparator (Self-Monitoring of Blood Glucose - SMBG) **Blinding:** Open-label **Randomization:** Randomized, Two-arm

3.2 Planned Enrollment & Duration Target \$N\$: 270 **Number of Sites:** Multi-center (14 locations) **Estimated Study Start:** 14-Apr-2025 **Estimated Primary Completion:** 15-Jun-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult subjects ≥ 18 years of age with Type 1 Diabetes mellitus (T1D) or Type 2 Diabetes mellitus (T2D) diagnosed at least 12 months prior to screening. 2. Using a multiple daily injection (MDI) regime for at least six months prior to screening, currently performing SMBG, with no CGM/flash glucose monitoring (FGM) use during the last six months. 3. HbA1c $\geq 8\%$ and $\leq 10\%$ based on analysis from a local laboratory.

4.2 Exclusion Criteria 1. Pregnancy (lack of negative pregnancy test), planned pregnancy, or breastfeeding. 2. Severe visual impairment, untreated adrenal or thyroid insufficiency, or significant renal impairment (eGFR <30 ml/min within last year). 3. Allergies to the adhesive (glue or tape), skin diseases at sensor application sites, or conditions making it unsafe to target HbA1c $<7\%$. 4. Taking interfering substances such as ≥ 300 mg hydroxyurea, levodopa, methyldopa, ascorbic acid, or anticipated acute uses of glucocorticoids.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Continuous sensor wear (Accu-Chek SmartGuide) for real-time interstitial glucose monitoring compared to daily standard frequency of fingerstick SMBG. **Route of Administration:** Subcutaneous sensor application. **Duration of Treatment:** 26 weeks

5.2 Concomitant & Prohibited Therapy Permitted: Prescribed multiple daily injection (MDI) insulin therapies. **Prohibited:** Routine use of non-protocol CGM devices, high-dose interfering agents (e.g., ascorbic acid, hydroxyurea), and high-altitude hiking or elective surgeries requiring general anesthesia/MRI during the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Day -14 to 0 Informed Consent, Medical History, HbA1c Labs, eGFR confirmation Baseline Day 0 Randomization, Initial CGM Placement / SMBG Supply, App Setup Interim Week 13 $\pm$ 7 Sensor Replacement Check, Hypoglycemia Audit, App Adherence Review End of Study Week 26 Final TIR Calculation, Sensor Removal, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Mean change from baseline in overall time in range (TIR) of blood glucose concentrations

defined as 70-180 mg/dL. **Measurement Method:** Aggregated interstitial glucose data exported from the Accu-Chek SmartGuide Predict app versus logbook values from the SMBG arm.

7.2 Secondary & Exploratory Endpoints Level 1 Hypoglycemia (glucose ≥ 54 to <70 mg/dL for at least 15 mins) and Level 2 Hypoglycemia (<54 mg/dL for ≥ 15 mins) rates. Overall change in HbA1c from baseline.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient-reported skin irritation, adhesive reactions, and sensor-site infections. **Serious Adverse Events (SAEs):** Severe hypoglycemic/hyperglycemic events requiring third-party assistance or hospitalization; reported within 24 hours. **Data Monitoring Committee (DMC):** Internal safety monitoring evaluating sensor malfunctions and acute metabolic events.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary TIR comparison; Safety Set for dermatological and device-related AEs. **Sample Size Justification:** Powered ($\alpha = 0.05$, $\text{Power} = 80\%$) to detect a statistically significant percentage increase in overall TIR. **Handling of Missing Data:** Multiple Imputation for fragmented CGM wear-time bridging; minimum required sensor active-time percentages applied.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular compliance tracking via SmartGuide App data synchronization. **Audit Trail:** E-logs of CGM reading transmissions to standard reporting dashboards and query logs for missing calibration points.

11. Study Identifiers & Governance

Primary ID: DC000129 **Registry Number:** NCT06704672 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** Clinical Study to Evaluate the Impact of the Accu-Chek SmartGuide CGM Solution on the Mean Change in Time in Range Compared With Self-Monitoring of Blood Glucose in Participants With Type 1 and Type 2 Diabetes Mellitus **Official Regulatory Title:** Clinical Study to Evaluate the Impact of the Accu-Chek SmartGuide CGM Solution on the Mean

12. Clinical Framework (Hyperkalemia Specific) Adapted to Trial Condition

Primary Condition: Type 1 and Type 2 Diabetes Mellitus **Diagnosis Threshold:** HbA1c $\geq 8\%$ and $\leq 10\%$ on MDI therapy **Medical Methodology:** Continuous Glucose Monitoring (CGM) vs. SMBG Management **Optimization Target:** Time in Range (TIR) between 70-180 mg/dl

13. Study Procedures & Methodology

Management Pathway: Multiple Daily Injection (MDI) regimen optimization informed by glycemic variability. **Education Protocol (HCP):** Integration of Accu-Chek SmartGuide Predict app data into clinical workflows. **Education Protocol (Patient):** Sensor application, SmartGuide app setup, interpretation of real-time alarms. **Quality Audit Mechanism:** Device wear-time adherence tracking and connectivity assessments.

14. Observation & Follow-Up Schedule

Total Duration: 26 weeks **Onsite Visit Cadence:** Device initialization, interim safety checks, and final endpoint capture. **Remote Monitoring Frequency:** Continuous ambient data collection via synchronized smartphone apps. **Checklist Review Interval:** Routine check-ins to prevent sensor fatigue and secure compliance.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	TBD	____	02-Apr-2026	Clinical Operations Lead	TBD	____	02-Apr-2026
TBD	____	02-Apr-2026		Lead Statistician			